USING DDL STATEMENTS TO CREATE AND MANAGE TABLES

1. Create table DEPT1 with the following columns: DEPTNO INTEGER PRIMARY KEY, DNAME VARCHAR2(30) NOT NULL, LOC VARCHAR2(30) NOT NULL.

SQL> create table dept1(deptno integer primary key,dname varchar2(30) not null, loc varchar2(30) not null);

```
SQL> create table dept1(deptno integer primary key,dname varchar2(30) not null, loc varchar2(30) not null);

Table created.

SQL> desc dept1;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DEPTNO                                    NOT NULL NUMBER(38)
 DNAME                                     NOT NULL VARCHAR2(30)
 LOC                                       NOT NULL VARCHAR2(30)

SQL> |
```

2. Create table EMP1 with the following columns: EMPNO INTEGER PRIMARY KEY, ENAME VARCHAR2(20) NOT NULL, SAL NUMBER(10,2) CHECK (SAL > 5000), MGR NUMBER (Foreign Key referencing EMP1.EMPNO), DEPTNO INTEGER (Foreign Key referencing DEPT1.DEPTNO).

SQL> create table emp1(empno integer primary key,ename varchar2(20) not null,sal number(10,2) check (sal>5000),mgr number,deptno integer, foreign key(deptno) references dept1(deptno));

```
SQL> create table emp1(empno integer primary key,ename varchar2(20) not null,sal number(10,2) check (sal>5000),mgr numbe
r,deptno integer, foreign key(deptno) references dept1(deptno));

Table created.

SQL> desc emp1;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                     NOT NULL NUMBER(38)
 ENAME                                     NOT NULL VARCHAR2(20)
 SAL                                                NUMBER(10,2)
 MGR                                                NUMBER
 DEPTNO                                             NUMBER(38)
```

3.Create tables DEPT11 and EMP11 by copying DEPT1 and EMP1.

SQL> create table dept11 as select * from dept1;

```
SQL> create table dept11 as select * from dept1;

Table created.

SQL> desc dept11;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DEPTNO                                             NUMBER(38)
 DNAME                                     NOT NULL VARCHAR2(30)
 LOC                                       NOT NULL VARCHAR2(30)
```

SQL> create table emp11 as select * from emp1;

```
SQL> create table emp11 as select * from emp1;

Table created.

SQL> desc emp11;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                              NUMBER(38)
 ENAME                                     NOT NULL VARCHAR2(20)
 SAL                                                NUMBER(10,2)
 MGR                                                NUMBER
 DEPTNO                                             NUMBER(38)
```

4.Add a new column ADDRESS VARCHAR2(30) to the EMP1 table.

SQL> alter table emp1 add Address varchar2(30);

```
SQL> alter table emp1 add Address varchar2(30);

Table altered.

SQL> desc emp1;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                     NOT NULL NUMBER(38)
 ENAME                                     NOT NULL VARCHAR2(20)
 SAL                                                NUMBER(10,2)
 MGR                                                NUMBER
 DEPTNO                                             NUMBER(38)
 ADDRESS                                            VARCHAR2(30)
```

5. Rename the SAL column in EMP1 to SALARY.

SQL> alter table emp1 rename column sal to salary;

```
SQL> alter table emp1 rename column sal to salary;

Table altered.

SQL> desc emp1;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                     NOT NULL NUMBER(38)
 ENAME                                     NOT NULL VARCHAR2(20)
 SALARY                                             NUMBER(10,2)
 MGR                                                NUMBER
 DEPTNO                                             NUMBER(38)
 ADDRESS                                            VARCHAR2(30)
```

6.Rename the foreign key constraint names in EMP1 and verify the constraint names.

```
SQL> SELECT constraint_name,
       constraint_type,
       table_name
FROM user_constraints
WHERE table_name = 'EMP1';
```

```
SQL> ALTER TABLE EMP1  RENAME CONSTRAINT SYS_C007845 TO FK_EMP1;
```

Table altered.

7.Increase the size of the ENAME column to VARCHAR2(40).

SQL> alter table emp1 modify ename varchar2(40);

```
SQL> alter table emp1 modify ename varchar2(40);

Table altered.

SQL> desc emp1;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                     NOT NULL NUMBER(38)
 ENAME                                     NOT NULL VARCHAR2(40)
 SALARY                                             NUMBER(10,2)
 MGR                                                NUMBER
 DEPTNO                                             NUMBER(38)
 ADDRESS                                            VARCHAR2(30)
```

8. Drop the NOT NULL constraint on the ENAME column.

SQL> ALTER TABLE emp1 MODIFY ename varchar2(40) NULL;

```
SQL> ALTER TABLE emp1 MODIFY ename varchar2(40) NULL;

Table altered.

SQL> desc emp1;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                     NOT NULL NUMBER(38)
 ENAME                                              VARCHAR2(40)
 SALARY                                             NUMBER(10,2)
 MGR                                                NUMBER
 DEPTNO                                             NUMBER(38)
 ADDRESS                                            VARCHAR2(30)

SQL>
```

9.Create a comment on the DEPT1 table as 'Depts of WIPRO'.

SQL> comment on table dept1 is 'Depts of WIPRO';

Comment created.

10.Create a comment on the DEPTNO column of DEPT1 as 'Deptno of WIPRO'. SQL> COMMENT ON COLUMN dept1.deptno is 'deptno of WIPRO';

Comment created.

11. Create a comment on the EMP1 table as 'Employees of WIPRO'.

SQL> comment on table emp1 is 'Employees of WIPRO';

Comment created.

12.Create a comment on the EMPNO column of EMP1 as 'Empno of WIPRO'.

SQL> COMMENT ON COLUMN emp11.empno is 'empno of WIPRO';

Comment created.

13.Remove all comments from the tables and columns.

SQL> COMMENT ON table emp1 IS ' ';

SQL> COMMENT ON COLUMN emp1.empno IS ' ';

SQL> COMMENT ON table dept1 IS ' ';

SQL> COMMENT ON COLUMN dept1.deptno IS ' ';

14.Mark the SALARY and ENAME columns of EMP1 as UNUSED.

ALTER TABLE emp1 SET UNUSED (salary,ename);

```
SQL> ALTER TABLE emp1  SET UNUSED (salary,ename);

Table altered.

SQL> desc emp1;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                     NOT NULL NUMBER(38)
 MGR                                                NUMBER
 DEPTNO                                             NUMBER(38)
 ADDRESS                                            VARCHAR2(30)

SQL>
```

15.Drop the UNUSED columns from EMP1.

ALTER TABLE emp1 DROP UNUSED COLUMNS;

16.Drop the EMP1 and DEPT1 tables.

SQL> drop table emp1;

Table dropped.

SQL> drop table dept1;

Table dropped.

```
SQL> drop table emp1;

Table dropped.

SQL> drop table dept1;

Table dropped.

SQL> desc emp1;
ERROR:
ORA-04043: object emp1 does not exist


SQL> desc dept1;
ERROR:
ORA-04043: object dept1 does not exist
```

17.Create table EMP1 from the EMP table by copying both the structure and data. Verify the data.

SQL> create table emp1 as select * from emp;

Table created.

```
SQL> create table emp1 as select * from emp;

Table created.

SQL> select * from emp1;

     EMPID ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTID
----------
      7839 KING       PRESIDENT            17-NOV-81       5000
        10

      7788 BLAKE      MANAGER         7839 01-MAY-81       2850
        30

      7799 Johnson    MANAGER         7342 01-JUN-87       2855
        30
```

18.Rename EMP1 to EMP_TEST.

SQL> alter table emp1 rename to emp_test;

Table altered.

```
SQL> alter table emp1 rename to emp_test;

Table altered.

SQL> select * from emp_test;

     EMPID ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTID
----------
      7839 KING       PRESIDENT            17-NOV-81       5000
        10

      7788 BLAKE      MANAGER         7839 01-MAY-81       2850
        30

      7799 Johnson    MANAGER         7342 01-JUN-87       2855
        30
```

19. Truncate the EMP_TEST table and verify that all rows have been deleted.
SQL> truncate table emp_test;

SQL> select * from emp_test;

```
SQL> truncate table emp_test;

Table truncated.

SQL> select * from emp_test;

no rows selected
```

20.Create table EMP2 from EMP by copying only the EMPNO, ENAME, and SAL columns along with their data.

SQL> create table emp2 as select empid,ename,sal from emp;

```
SQL> create table emp2 as select empid,ename,sal from emp;

Table created.

SQL> desc emp2;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPID                                              NUMBER(4)
 ENAME                                              VARCHAR2(10)
 SAL                                                NUMBER(7,2)
```

21.Drop the EMP2 table.

SQL> drop table emp2;

Table dropped.

```
SQL> drop table emp2;

Table dropped.

SQL> desc emp2;
ERROR:
ORA-04043: object emp2 does not exist
```

22.Create the EMP2 table again from EMP without copying any data.

SQL> CREATE TABLE emp2   AS   SELECT * FROM emp WHERE 1=0;

Table created.

SQL> desc emp2;

```
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPID                                              NUMBER(4)
 ENAME                                              VARCHAR2(10)
 JOB                                                VARCHAR2(9)
 MGR                                                NUMBER(4)
 HIREDATE                                           DATE
 SAL                                                NUMBER(7,2)
 COMM                                               NUMBER(7,2)
```

DEPTID NUMBER(2)

SQL> select * from emp2;

no rows selected

23.Drop the EMP2 table.

SQL> drop table emp2;

Table dropped.

24.Flashback the first dropped EMP2 table (containing data) and verify the table.

SQL> FLASHBACK TABLE emp2 TO BEFORE DROP;

```
SQL> SHOW RECYCLEBIN;
ORIGINAL NAME    RECYCLEBIN NAME                OBJECT TYPE  DROP TIME
---------------- ------------------------------ ------------ -------------------
DEPT             BIN$pTu+QthxRDurrePEUZZTWA==$0 TABLE        2024-09-12:10:21:56
DEPT1            BIN$YvNpjWiZS4Gu3dykRlS+/Q==$0 TABLE        2024-09-28:14:25:04
DEPT1            BIN$AjO2V40eRC+/rPLFTKIaoQ==$0 TABLE        2024-09-21:11:49:11
DEPT11           BIN$17Wv73TcRZWYb0rhj85+1A==$0 TABLE        2024-09-21:11:59:53
EMP              BIN$oJI9pKsFQC6EOdPMIvWrLA==$0 TABLE        2024-09-12:10:21:49
EMP1             BIN$Dxs3bh7vSoOUvq+EmjQ1Nw==$0 TABLE        2024-09-28:14:31:22
EMP1             BIN$b5ai/y0bQ5ScvsrR7oVkOQ==$0 TABLE        2024-09-28:14:24:57
EMP2             BIN$DwsV7foFQAqQpx3UFbODLA==$0 TABLE        2024-09-30:11:17:35
EMP2             BIN$9dyGPwVPSg2LvVDuZTHzHA==$0 TABLE        2024-09-30:11:06:21
EMP2             BIN$fhw62iV8Sy2SapAZK+EPQQ==$0 TABLE        2024-09-30:10:57:21
SQL> FLASHBACK TABLE emp2 TO BEFORE DROP;

Flashback complete.

SQL> select * from emp2;

     EMPID ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTID
----------
      7839 KING       PRESIDENT            17-NOV-81       5000
        10

      7788 BLAKE      MANAGER         7839 01-MAY-81       2850
        30
```

25.Flashback the recently dropped table and restore it as EMP2_1.

SQL> FLASHBACK TABLE emp2 TO BEFORE DROP RENAME TO emp2_1;

Flashback complete.

```
SQL> FLASHBACK TABLE emp2 TO BEFORE DROP RENAME TO emp2_1;

Flashback complete.

SQL> select * from emp2_1;

     EMPID ENAME      JOB              MGR HIREDATE         SAL       COMM
---------- ---------- --------- ---------- --------- ---------- ----------
    DEPTID
----------
      7839 KING       PRESIDENT            17-NOV-81       5000
        10

      7788 BLAKE      MANAGER         7839 01-MAY-81       2850
        30

      7799 Johnson    MANAGER         7342 01-JUN-87       2855
        30
```

26.Display all tables owned by the current user.

```
SQL> SELECT table_name  FROM user_tables;

TABLE_NAME
------------------------------
REGIONS
LOCATIONS
DEPARTMENTS
JOBS
EMPLOYEES
JOB_HISTORY
DEPT
EMP
LODGE_DETAILS
SKILL_DETAILS
EMP_DETAILS

TABLE_NAME
------------------------------
EMP_SKILL
DEPT11
EMP11
STUDENT
SALGRADE
DEPT_1
EMP_1
PRODUCTSTB
DEPT1
EMP2_1
```

```
COUNTRIES

TABLE_NAME
------------------------------
EMP_TEST

23 rows selected.
```